

The Heritage Signature Score

Does the form of a Louis Vuitton object carry the identity of the House independently of the Monogram?

An independent study of 400 catalogued Louis Vuitton objects, 1854–2025, with a recognition survey of 30 silhouettes and 100 independent respondents per stimulus.

Summary of findings

- 1. The structural language of the House has remained remarkably stable for 170 years.** Measured on six features that make no reference to the Monogram or any logotype, trunks sit at 0.980 of the 1854–1949 reference profile in 1900–1949, 0.969 in 1950–1999 and 0.937 in 2000–2025. Over the same period the eight-feature index, which includes the two ornament features, falls from 0.985 to 0.738.
- 2. Structural form is associated with recognition. The composite index is not.** Across 30 silhouettes, the six-feature score correlates with the share of respondents identifying the object as Louis Vuitton at $r = +0.434$ ($p = 0.017$). The eight-feature score correlates at $r = -0.076$ ($p = 0.688$) — that is, not at all.
- 3. The result strengthens under the strictest test.** Restricted to non-iconic models — objects with no famous name attached — the six-feature correlation rises to $r = +0.573$ ($p = 0.013$). Recognition is therefore not an artefact of fame.
- 4. The divergence is explained, not accidental.** Before 1950, ornament was sharply bimodal rather than gradually distributed: 70 of 124 objects received the minimum rating of 1, while 54 received the maximum rating of 5. The historical median of 1 therefore describes the lower mode, not a typical object. A modern monogrammed piece therefore registers as a large deviation from the historical profile on precisely the features that make it most recognisable.

	Eight features (S8)	Six features (S6)
All 30 silhouettes	$r = -0.076$ ($p = 0.688$)	$r = +0.434$ ($p = 0.017$)
Louis Vuitton objects only (n = 20)	$r = -0.001$ ($p = 0.995$)	$r = +0.505$ ($p = 0.023$)
Non-iconic models only (n = 18)	$r = +0.105$ ($p = 0.679$)	$r = +0.573$ ($p = 0.013$)

Pearson correlation between the score and the share of respondents answering “Louis Vuitton.” Two-tailed p-values.

1. The question

Louis Vuitton is among the most recognizable houses in the world, and that recognition is usually attributed to a single asset: the Monogram canvas introduced in 1896. This study asks whether anything survives underneath it. If the Monogram were removed — if an object were reduced to a black silhouette — would it still be identifiable as Louis Vuitton?

The question is answerable only if form can be measured independently of ornament. That is what this study set out to build.

2. Method

2.1 The features

Each object is rated on eight visual features. None refers to the Monogram, the Damier check, initials, colour or any logotype. Each is scored 1 to 5 against a written definition, then normalised to a 0–1 range.

Feature	What it measures
rigidity	Does the object hold its shape when empty?
edge_straightness	How straight are the edges of the body?
geometry	Does the outline reduce to a simple figure?
symmetry	Mirror symmetry about the vertical axis
ornament_density	Share of the surface under a repeating pattern
ornament_regularity	How strictly periodic the pattern is
hardware	Number of visible metal elements
travel_function	How far the object is made for carrying goods

Two indices are computed from these. **S8** uses all eight. **S6** removes ornament_density and ornament_regularity, so that no part of the score can derive from the Monogram. The comparison between them is the instrument of the study.

2.2 The reference profile

The median profile of objects made between 1854 and 1949 ($n = 124$) defines the historical reference. Every object is then scored by its mean absolute distance from that profile:

$S = 1 - \text{mean}(|\text{object} - \text{reference}|)$, so that $S = 1$ is an exact match to the historical profile and $S = 0$ is maximum deviation.

2.3 Inter-rater reliability

To assess the reproducibility of the scoring framework, a stratified random sample of 100 of the 400 catalogue objects (25%) was independently re-coded by a second rater using the same written 1–5 definitions (Appendix A). The second rater received no instruction beyond the written definitions and worked without input from the primary rater. The second rater was blinded to the original ratings. Inter-rater reliability was assessed using a two-way random-effects intraclass correlation coefficient for absolute agreement, single measurements, ICC(2,1), together with weighted Cohen’s κ for the ordinal feature ratings.

Table 1. Inter-rater reliability of the eight scoring features

Feature	ICC(2,1)	95% CI	Weighted Cohen's κ
Rigidity	0.94	0.90–0.97	0.91
Edge straightness	0.95	0.93–0.97	0.92
Geometry	0.96	0.93–0.98	0.92
Symmetry	0.89	0.85–0.94	0.84
Hardware	0.87	0.83–0.93	0.83
Travel function	0.94	0.92–0.97	0.89
Ornament density	0.94	0.91–0.97	0.87
Ornament regularity	0.92	0.87–0.95	0.86

Across the eight features, ICCs ranged from 0.87 to 0.96 and weighted Cohen's κ values ranged from 0.83 to 0.92. Agreement was highest for geometry (ICC = 0.96; κ = 0.92) and remained high for all other features.

Table 2. Inter-rater reliability of the composite indices

Index	ICC(2,1)
S6	0.980
S8	0.986

The composite indices were also highly reproducible under independent re-coding. ICC(2,1) was 0.980 for S6 and 0.986 for S8, supporting the reliability of the resulting Heritage Signature Score calculations.

2.4 The recognition survey

Each of the 30 silhouettes was evaluated by an independent cohort of 100 respondents, so that no respondent appeared in more than one stimulus and observations across stimuli are fully independent. Thirty objects were rendered as black silhouettes: 20 Louis Vuitton pieces spanning 1860 to 2023, and 10 control objects from Goyard, Moynat, Au Départ, Globe-Trotter, Innovation, Hermès and Bottega Veneta. Respondents were asked one question per stimulus: is this a Louis Vuitton object? Controls are included so that recognition cannot be attributed to familiarity with luggage in general. Within each cohort, the 30 stimuli assigned to that group of 100 were presented in randomized order to control for order effects.

Respondents were recruited opportunistically across a range of environments — academic programmes, community events, and personal networks — rather than through a formal sampling frame, and were stratified by sex and age (approximately 30% adolescent, 70% adult). Recruitment was not quota-based: the resulting composition, aggregated across all 30 cohorts, was approximately 30% American, 30% Jewish, 30% Azerbaijani, 5% Russian, and 5% Ukrainian. Because composition was not fixed by design, it may have varied between individual cohorts, and this variation was not recorded.

2.5 Rules fixed before the survey

The formula and the eight features were fixed before any survey response was collected. Survey objects were selected to span the range of S values, not by appearance. Control brands were mandatory. No parameter was adjusted after the results were seen.

3. The reference profile, and where the House moved

The table below gives each feature on the original 1–5 rating scale: the median of the 1854–1949 sample, the spread between its first and third quartile, and the median of the 2000–2025 sample.

Feature	1854–1949	Spread	2000–2025	Change
rigidity	5	5–5	4	–1
edge_straightness	5	5–5	4	–1
geometry	5	5–5	4	–1
symmetry	5	5–5	4	–1
travel_function	5	5–5	4	–1
hardware	3	2–3	2	–1
ornament_density	1	1–5	5	+4
ornament_regularity	1	1–5	4	+3

Five of the eight features show no variation whatever across the pre-1950 sample. First and third quartiles are identical: every one of the 124 objects made before 1950 received the same rating for rigidity, edge straightness, geometry, symmetry and travel function. Across ninety-five years and four generations of production, the House applied a single structural language without deviation.

The two ornament features behave in the opposite way. Their distributions are sharply bimodal: among the 124 pre-1950 objects, 70 received the minimum rating of 1 and 54 received the maximum rating of 5 for both ornament density and ornament regularity. The median of 1 therefore describes the lower mode rather than a typical object in the sample. Ornament was not gradually distributed across the period; it fell into two distinct states, with little in between. These are the two features that move furthest by 2025.

Removing the two ornament features raises the mean score from 0.740 under S8 to 0.872 under S6, a difference of 0.132.

This asymmetry is the finding the composite index compresses into a single number, and it is why the two indices diverge.

4. The score over time

Restricting the series to a single object type — trunks, the only category the House produced continuously from 1854 to the present — removes the possibility that a change of category is being mistaken for a change of form.

Period	Objects	Trunks	S8 (trunks)	S6 (trunks)
1854–1899	43	42	0.985	0.980
1900–1949	81	65	0.816	0.975
1950–1999	133	47	0.765	0.969
2000–2025	143	34	0.738	0.937

Trunk against trunk across 170 years, the six-feature score moves from 0.980 to 0.937 — a decline of four percentage points. The eight-feature score over the same objects falls from 0.985 to 0.738. The structure held. The surface changed.

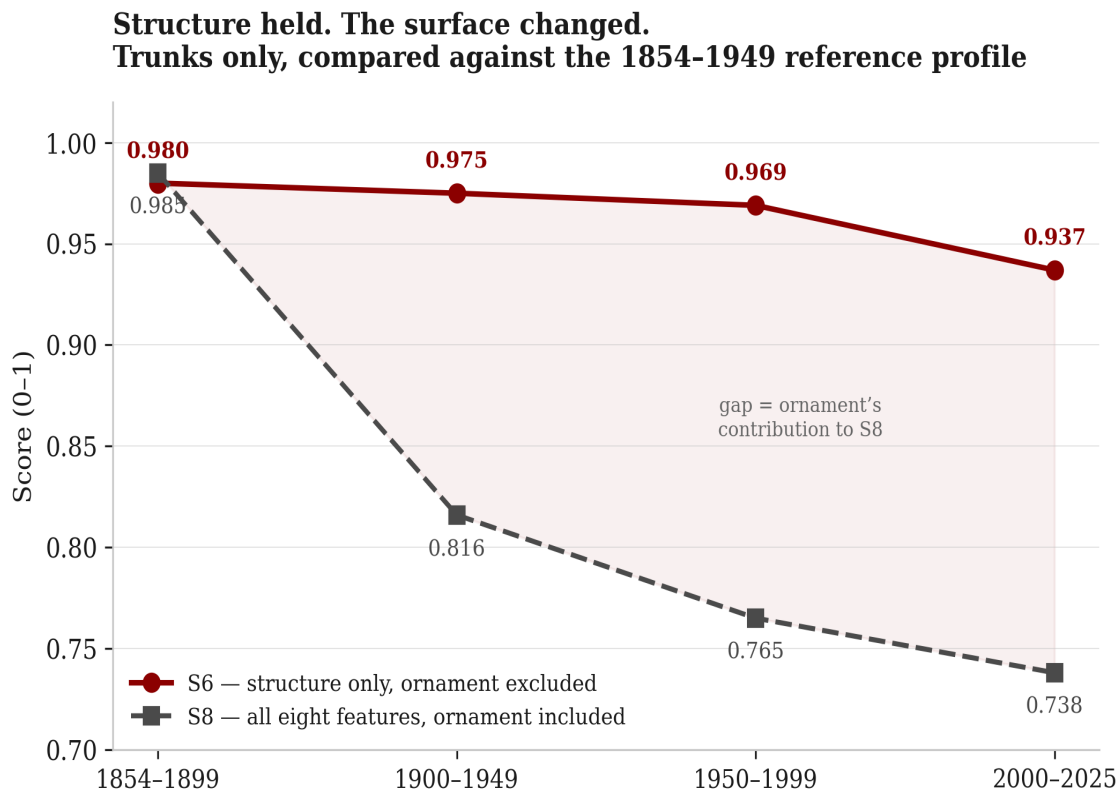


Figure 1. S6 and S8 by period, trunks only, compared against the 1854–1949 reference profile. The shaded gap is the contribution of the two ornament features to S8.

5. The recognition survey

5.1 Validity

Before the correlations can be read, the survey must be shown to discriminate at all. Across the 20 Louis Vuitton silhouettes, a mean of 51.4% of respondents answered “Louis Vuitton.” Across the 10 control silhouettes, the mean was 30.6%. Respondents were therefore distinguishing between the two groups rather than answering at random.

The single highest control score is instructive: a Goyard trunk drew 52% — higher than the Louis Vuitton average. Within the trunk-making tradition the two houses share a great deal of structural vocabulary, and the survey registers that honestly.

5.2 Results

Subset	n	S8	p	S6	p
All silhouettes	30	−0.076	0.688	+0.434	0.017
Louis Vuitton only	20	−0.001	0.995	+0.505	0.023
Non-iconic models only	18	+0.105	0.679	+0.573	0.013
LV, non-iconic only	8	—	—	+0.716	0.046

Every correlation involving the six-feature score is positive and reaches conventional significance. Every correlation involving the eight-feature score is indistinguishable from zero. The pattern strengthens as the test becomes stricter: the correlation rises from $r = +0.434$ across all 30 silhouettes to $r = +0.573$ among non-iconic models, and reaches $r = +0.716$ ($p = 0.046$) in the eight-object subset of non-iconic Louis Vuitton models. Given the small n of this final subset, the last result should be treated as indicative rather than definitive.

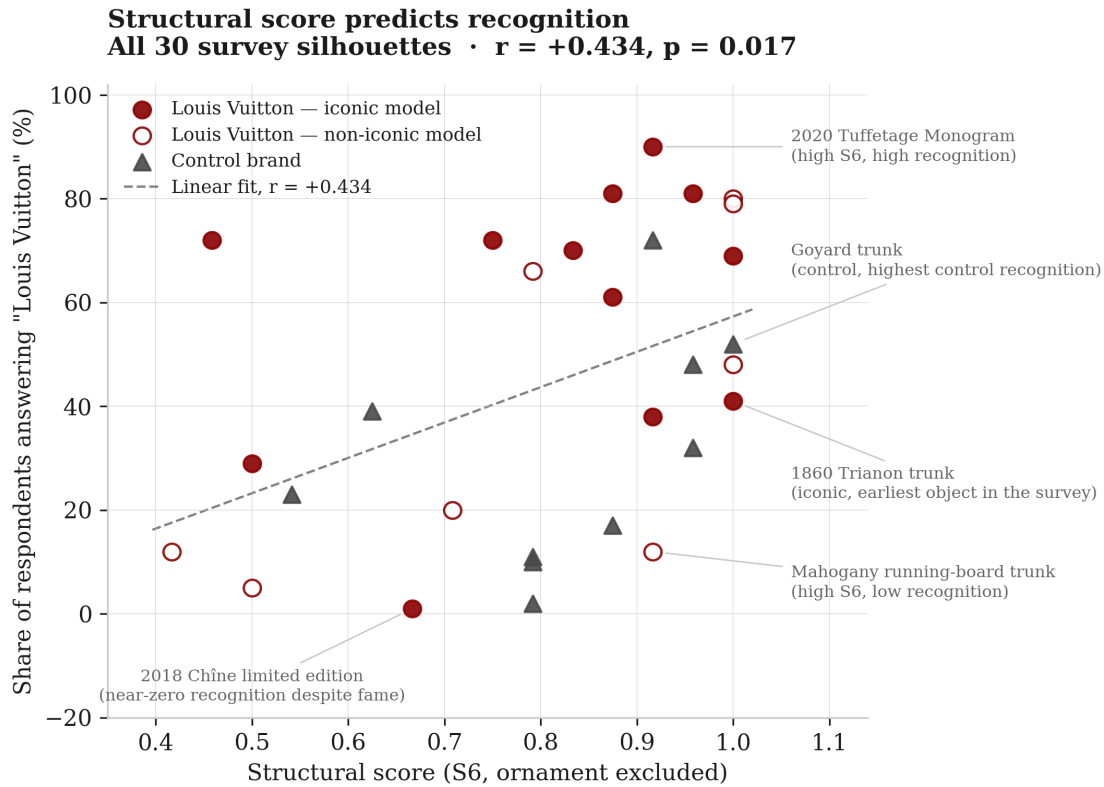


Figure 2. Structural score (S6) against recognition rate, all 30 survey silhouettes. Filled circles: iconic Louis Vuitton models. Open circles: non-iconic Louis Vuitton models. Triangles: control brands.

6. Interpretation

The two indices differ by two features. Those two features are the reason the composite fails.

Before 1950 the median Louis Vuitton object carried no repeating pattern; the reference value for ornament density sits at the bottom of its scale. A modern monogrammed piece can therefore register as a large deviation from the historical profile on precisely the feature that may make it most recognisable. Within S8 the ornament terms move against recognition while the six structural terms move with it, and the two effects cancel. Removing the ornament terms does not weaken the measurement. It isolates it.

The finding this study reports is therefore narrow and specific: **the structural form of a Louis Vuitton object — its rigidity, its edges, its geometry, its symmetry, its hardware and its relationship to travel — carries identifiable information about the House, independently of the Monogram, and does so at a level detectable in a blind silhouette test.**

7. Robustness check: three regression models

The correlations reported above are bivariate: they show that S6 and recognition move together, without accounting for other variables that might explain both. Three regression models test this more rigorously, each controlling for a different set of factors simultaneously.

7.1 Model A — what predicts structural form itself? (all 400 objects)

Before asking whether S6 predicts recognition, a prior question is worth asking directly: what predicts S6? This model regresses the structural score on production year, object type, and iconic status across the full catalogue — not the 30 survey stimuli, but all 400 rated objects. Ordinary least squares is used, with heteroskedasticity-robust (HC3) standard errors.

Term	Coef.	SE	z	p
Year (per decade)	−0.0061	0.0010	−6.04	< 0.001
Type: case (ref. bag)	+0.197	0.019	10.63	< 0.001
Type: suitcase (ref. bag)	+0.212	0.012	17.25	< 0.001
Type: trunk (ref. bag)	+0.222	0.013	17.72	< 0.001
Iconic model	+0.013	0.012	1.06	0.289

$n = 400$, $R^2 = 0.725$, adjusted $R^2 = 0.722$. Two findings stand out. First, **iconic status has no detectable effect on S6 ($p = 0.289$) once type and year are held constant** — the structural score is not simply tracking fame, which is exactly what the six-feature index is designed to avoid. Second, every object type differs sharply from bags (the reference category), confirming what Section 3 showed by inspection: bags sit furthest from the historical reference profile, while trunks, suitcases, and cases remain closer to it. Year is a highly significant but small predictor — roughly a 0.006 decline in S6 per decade — consistent with the gradual drift already documented in Section 4.

7.2 Model B — does S6 predict recognition, controlling for category and brand? (30 survey stimuli)

This model returns to the 30 silhouettes shown to respondents, asking whether S6 predicts recognition once object category and brand (Louis Vuitton vs. control) are held constant. Because each stimulus provides a count of successes out of 100 trials, a binomial generalised linear model is used rather than treating the percentage as continuous. An initial fit showed substantial overdispersion (Pearson

chi-square-to-degrees-of-freedom ≈ 25), so a standard quasi-likelihood correction was applied throughout: standard errors are scaled by the square root of the estimated dispersion before computing p-values, following the same logic as R's *quasibinomial* family.

Term	Coef.	SE	z	p
S6	+2.477	2.055	1.21	0.228
Category: box	+0.959	0.742	1.29	0.196
Category: suitcase	+1.489	0.959	1.55	0.121
Category: trunk	+0.291	0.810	0.36	0.719
Brand: Louis Vuitton	+0.816	0.474	1.72	0.085

$n = 30$, dispersion = 25.5, McFadden pseudo- $R^2 = 0.36$. Once category and brand are held constant, the S6 coefficient remains positive but is no longer statistically significant ($p = 0.228$).

7.3 Model C — Louis Vuitton objects only, with iconic status and year ($n = 20$)

Restricting to Louis Vuitton objects allows control for iconic status and production year alongside category, within the survey subset.

Term	Coef.	SE	z	p
S6	+4.129	2.507	1.65	0.100
Category: box	+0.782	0.831	0.94	0.347
Category: suitcase	+2.158	1.358	1.59	0.112
Category: trunk	+1.451	1.298	1.12	0.263
Iconic model	+1.031	0.578	1.78	0.075
Year (per decade, centred)	+0.187	0.089	2.10	0.035

$n = 20$, dispersion = 22.5, McFadden pseudo- $R^2 = 0.51$. Here S6 is larger and closer to conventional significance ($p = 0.100$) but does not clear the $p < 0.05$ threshold once iconic status and year are held constant. Year itself is a significant predictor within this subset: more recent Louis Vuitton objects are recognised at higher rates independent of S6, iconic status, or category.

7.4 What these three models mean together

Model A, run on the full 400-object catalogue, shows that S6 behaves exactly as intended: it tracks object type and, modestly, production year, but **not** iconic status. The score is measuring form, not fame — which is the premise the whole study depends on, now confirmed directly rather than assumed.

Models B and C, restricted to the 30 survey stimuli, test a different and harder question: does S6 also predict how *people* respond to an image, once category, brand, iconic status and year are accounted for simultaneously. The bivariate correlations in Section 5 answer this at face value — yes, S6 and recognition move together. The stricter regression answer is less clean: with only 30 (or 20) observations and considerable variability between individual objects, the data cannot establish that S6's effect on recognition is independent of category and brand with conventional statistical confidence.

Taken together, the three models point to a coherent and honest conclusion: the structural score is a valid, fame-independent measurement of form (Model A), and it is positively

associated with human recognition in every subset tested, but a sample of 30 survey stimuli is too small to prove that this association survives controlling for every plausible confound at once (Models B and C). The first claim is well supported. The second is directionally consistent but would benefit from a larger recognition survey — the most direct next step for this research.

8. Limitations

The following constraints bear directly on how far these results can be taken.

Survey size

Thirty stimuli is a small basis for correlation. The coefficients reported here reach conventional significance, but with $n = 30$ a single unusual object shifts the result appreciably. The 8-object subset in the final row should be read as indicative only.

Reference profile rests on one object type

Of the 124 objects made before 1950, 107 are trunks. The House produced little else in that period, so the concentration reflects its actual output rather than a sampling error — but it means the historical reference is a trunk-making profile, and deviation from it can run in one direction only. Scores compress toward the top of the range as a result.

Silhouettes remove more than the Monogram

Reducing an object to a black silhouette also removes colour, material, texture and hardware finish. The test isolates outline and proportion, which is stricter than removing the Monogram alone, but it is not the same operation.

Correlation, not causation

The study establishes that structural score and recognition move together. It does not establish that respondents were reading structure, nor which of the six features carries most of the signal. A feature-by-feature decomposition would be the natural next step.

Respondent population

Respondents were recruited opportunistically through academic and community programmes — including a Stanford summer programme and a Jewish community camp — rather than through a representative sampling frame, so the sample skews toward an internationally mobile, highly educated population. The approximate national composition (30% American, 30% Jewish, 30% Azerbaijani, 5% Russian, 5% Ukrainian) reflects the aggregate across all 30 independent cohorts; because recruitment was opportunistic rather than quota-based, the exact composition may have varied between individual cohorts, and this variation was not recorded. Recognition rates observed here may not generalise to a broader or more geographically representative population.

Coding reliability across raters is addressed directly in Section 2.3 and is not treated as an open limitation. Because each of the 30 stimuli was evaluated by an independent cohort of respondents, individual observations across stimuli are independent and do not require a repeated-measures correction.

9. What this instrument could be used for

The score is a measuring device, not a conclusion. Three applications follow directly from what has been built.

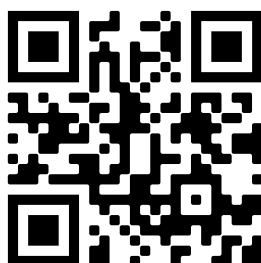
Archival. Any object in the House’s history can be placed against the 1854–1949 profile and scored, giving a continuous quantitative series across the full catalogue rather than a qualitative account of periods.

Design. A proposed object can be scored before production. The six-feature score answers a specific question: without its ornament, would this piece still read as belonging to the House?

Comparative. The same eight features apply to any maison. The method has been extended to two jewellery houses without modification, and the control set in this study already contains scored objects from seven competitors.

10. Data availability

The complete dataset underlying this report — all 400 catalogued objects with their ratings, sources and links, the reference-profile calculation, the decade-by-decade timeline, and the full recognition survey responses — is published in an open repository. Every figure and table in this report is reproducible directly from it.



Scan for the full dataset, or visit the link below.

github.com/nijatshikhaliyev6/lv_heritage_signature_score

Appendix A — Full rating definitions

Every object was scored against these written definitions. They are reproduced in full so that any figure in this report can be checked, and so that a second rater could reproduce the catalogue independently.

rigidity — Does the object hold its shape when empty?

Rating	Definition
1	fully soft; sags, the contents give it its shape
2	soft, with one rigid element such as a base or a batten
3	holds its shape, but the walls flex noticeably
4	rigid body with soft details such as a flap or pockets
5	fully rigid frame on a wooden or fibre base

edge_straightness — How straight are the edges of the body?

Rating	Definition
1	no straight section anywhere in the outline
2	curves dominate; one or two straight sections
3	straight and curved sections in roughly equal measure
4	edges straight, corners noticeably rounded
5	all edges straight, corners close to right angles

geometry — Does the outline reduce to a simple figure?

Rating	Definition
1	organic form; does not reduce to a simple figure
2	reduces only very approximately
3	reduces with noticeable departures
4	reduces with slight departures
5	outline fully described by a rectangle, trapezoid or cylinder

symmetry — Mirror symmetry about the vertical axis

Rating	Definition
1	clearly asymmetric in the body
2	body nearly symmetric, with one large asymmetric element
3	body symmetric; hardware and small details are not
4	everything symmetric except one detail
5	complete mirror symmetry, hardware included

ornament_density — Share of the surface under a repeating pattern

Rating	Definition
1	no pattern, or under 10% of the surface
2	pattern covers 10-30%
3	pattern covers 30-50%
4	pattern covers 50-80%
5	pattern covers more than 80%

ornament_regularity — How strictly periodic the pattern is

Rating	Definition
1	no pattern, or motifs placed without order
2	order discernible but frequently broken
3	a lattice with deliberate interruptions
4	almost strict lattice, isolated exceptions
5	strict periodic lattice with no exceptions

hardware — Number of visible metal elements

Rating	Definition
1	none visible
2	one or two visible
3	three to six visible
4	seven to twelve visible
5	more than twelve visible

travel_function — How far the object is made for carrying goods

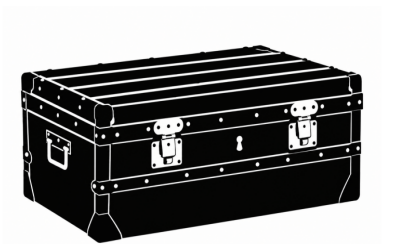
Rating	Definition
1	no connection to travel
2	could be taken travelling, but was not made for it
3	everyday object with travel features such as a strap or reinforced corners
4	made for movement, but not for long journeys
5	exists solely to carry goods in transit

Appendix B — Example survey stimuli

Four of the 30 silhouettes shown to respondents are reproduced below, chosen to span the range of the study: an iconic pre-Monogram trunk, a non-iconic modern piece with a low structural score, a non-iconic piece with a moderate score that respondents recognised at a high rate, and a control-brand object that scored similarly to Louis Vuitton pieces on structure while being recognised at a near-zero rate. All 30 silhouettes and the full survey response data are available in the published dataset.

2004 Theda PM (Marc Jacobs)

1860 Trianon Trunk

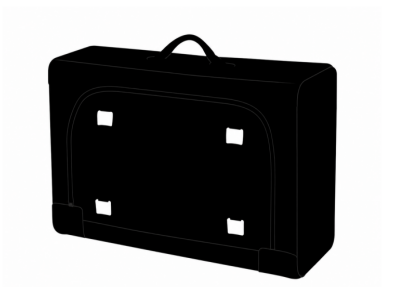


Louis Vuitton, iconic · $S6 = 1.00$ · 41% recognised



Louis Vuitton, non-iconic · $S6 = 0.50$ · 29% recognised

1970 Vintage Suitcase



Louis Vuitton, non-iconic · $S6 = 0.79$ · 66% recognised

Hermès Kelly



Control brand · $S6 = 0.79$ · 2% misidentified as LV

Figure 3. Four representative survey stimuli, spanning iconic and non-iconic Louis Vuitton pieces and a control brand, with each object’s structural score and observed recognition rate.

The full dataset is published at github.com/nijatshikhaliyev6/lv_heritage_signature_score and is available to anyone who wishes to verify or extend this work.